Ambient Intelligence-

Calming, Enriching and Empowering our lives

what **wiki** says is

> **...ambient intelligence (AmI) refers to electronic environments that are sensitive and responsive to the presence of people.**



What is AmI?

Ambient intelligence(AmI) is an emerging discipline that brings intelligence to our every-day environments and makes those environments sensitive to us.

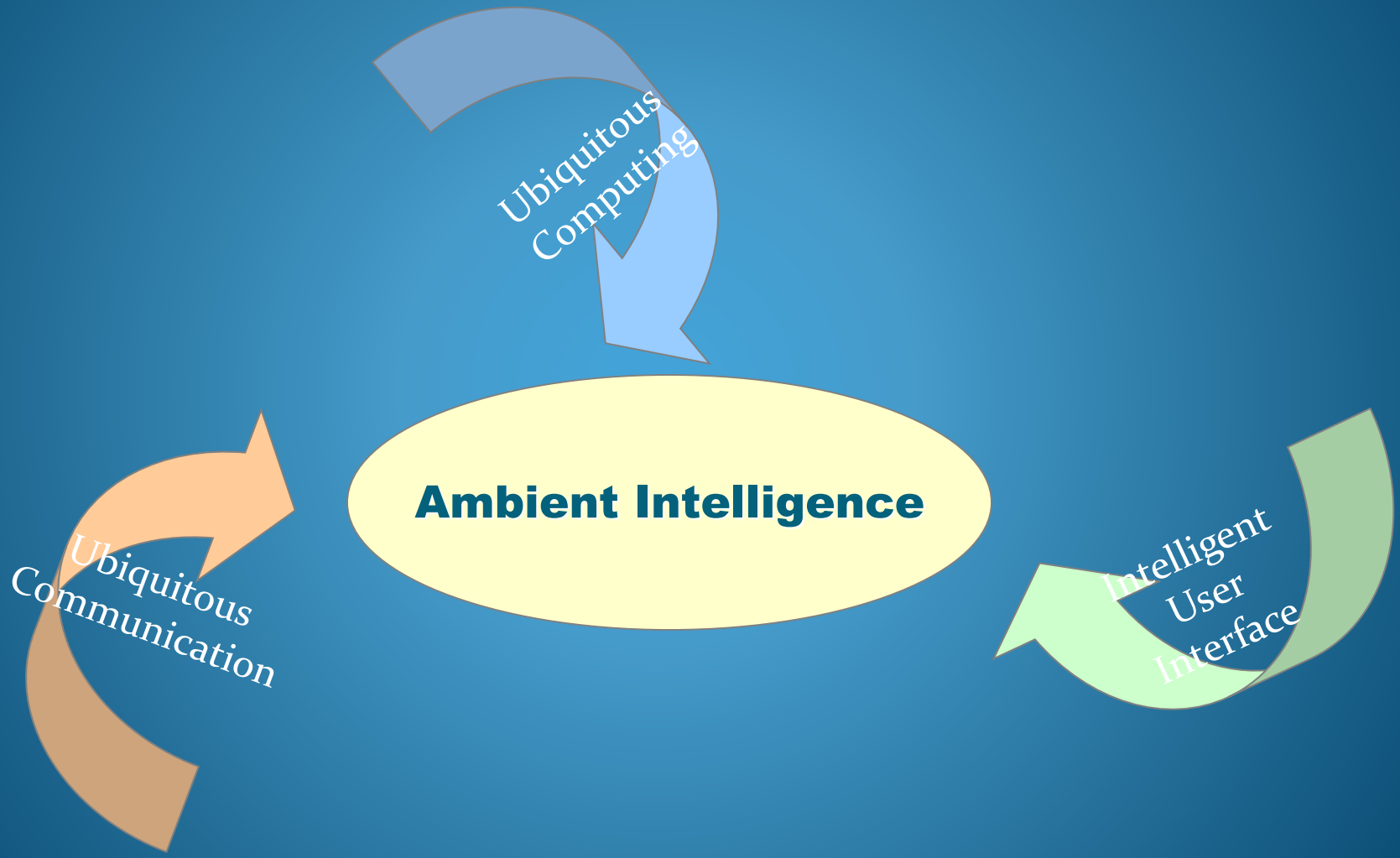
Ambient Intelligence is a network of hidden intelligent interfaces that recognize our presence and mould our environment to our immediate needs.

What is AmI?

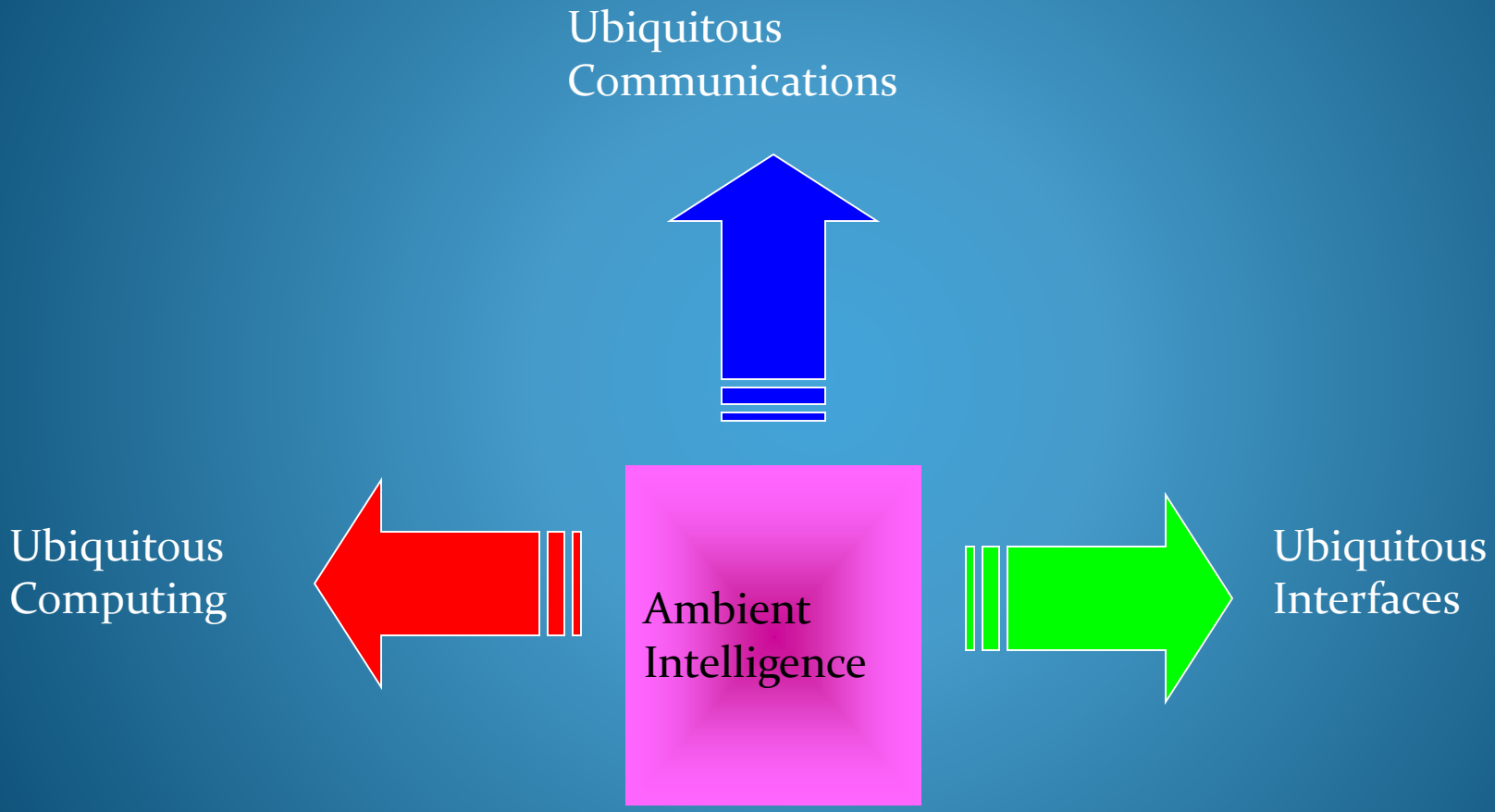
AmI refers to an exciting new paradigm in information technology, in which people are empowered through a **digital environment** that is aware of their presence and context and is **sensitive, adaptive and responsive** to their **needs, habits, gestures and emotions**.

It reduces the stress leading to an overall **higher quality of life.**

Key Technologies

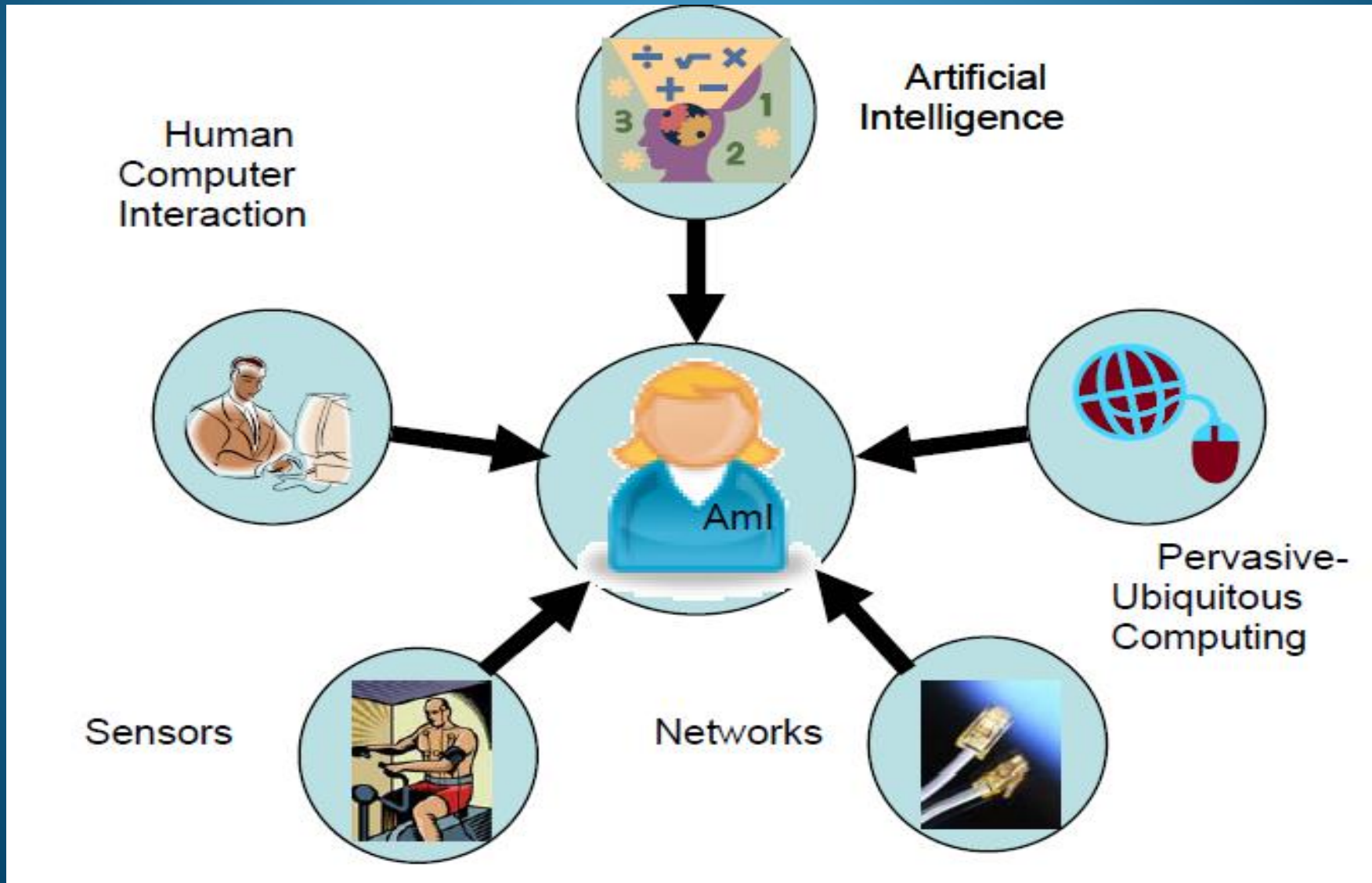


Convergence of 3 areas





Interaction in-between AmI and other disciplines



AmI


existing tools
context aware
adaptive
anticipatory
personalized
embedded

Many invisible dedicated devices throughout the environment.



embedded networked devices are integrated into the environment, that forms the core basis of ambient intelligence.

adaptive intelligence is a concept of changing response after acknowledging a change in us.

Adaptation to user involves understanding- Who the user is

- **User's particular needs, desires and habits.**

Environment changes constantly

-Devices must track and adapt



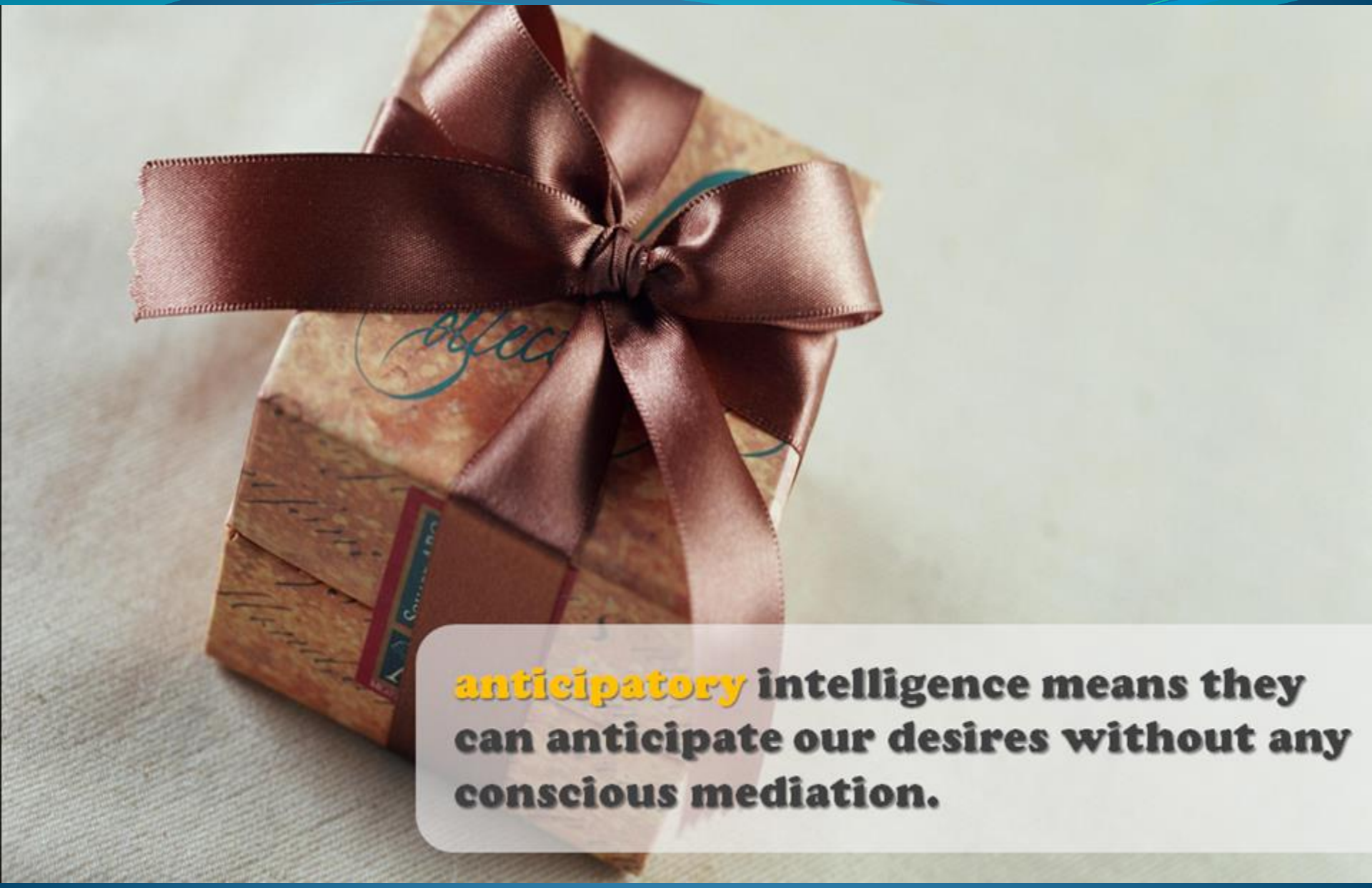
personalization is the concept of tailoring the technology according to our needs.

Personalization: learning your *explicit* and *implicit* needs



context aware devices can recognize us and our situational context.




anticipatory intelligence means they can anticipate our desires without any conscious mediation.

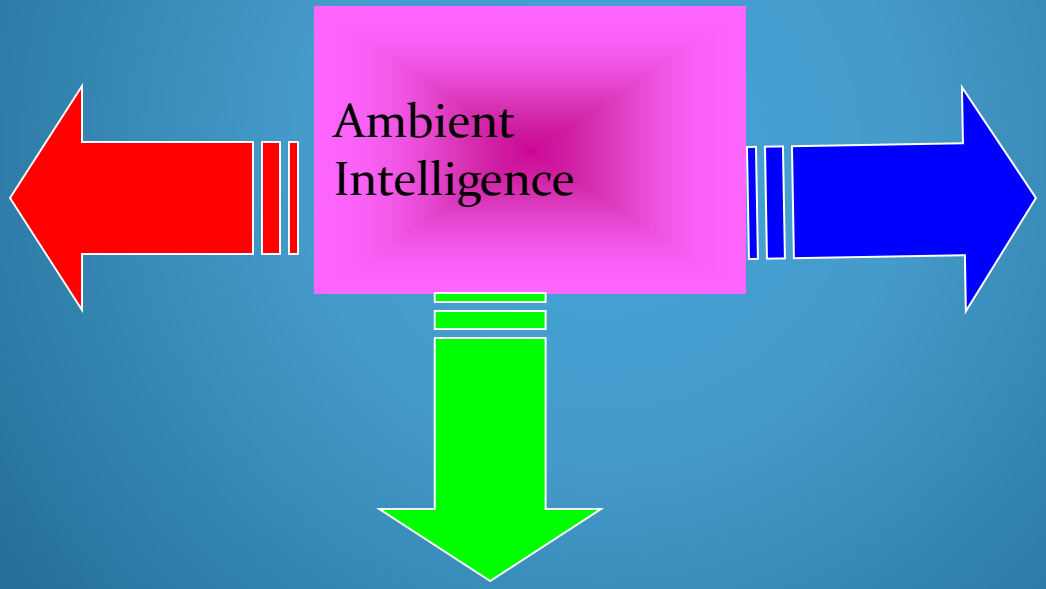
In SHORT

*.....is a vision of our future with disappearing hardware

Requirements of Ambient Intelligence

Unobtrusive hardware
- Nano-electronics & Nano-devices

Seamless communications infrastructure
- Heterogeneous networks
- Convergence networks



Dynamic and distributed device network
- **Portable**
- Plug-and-play

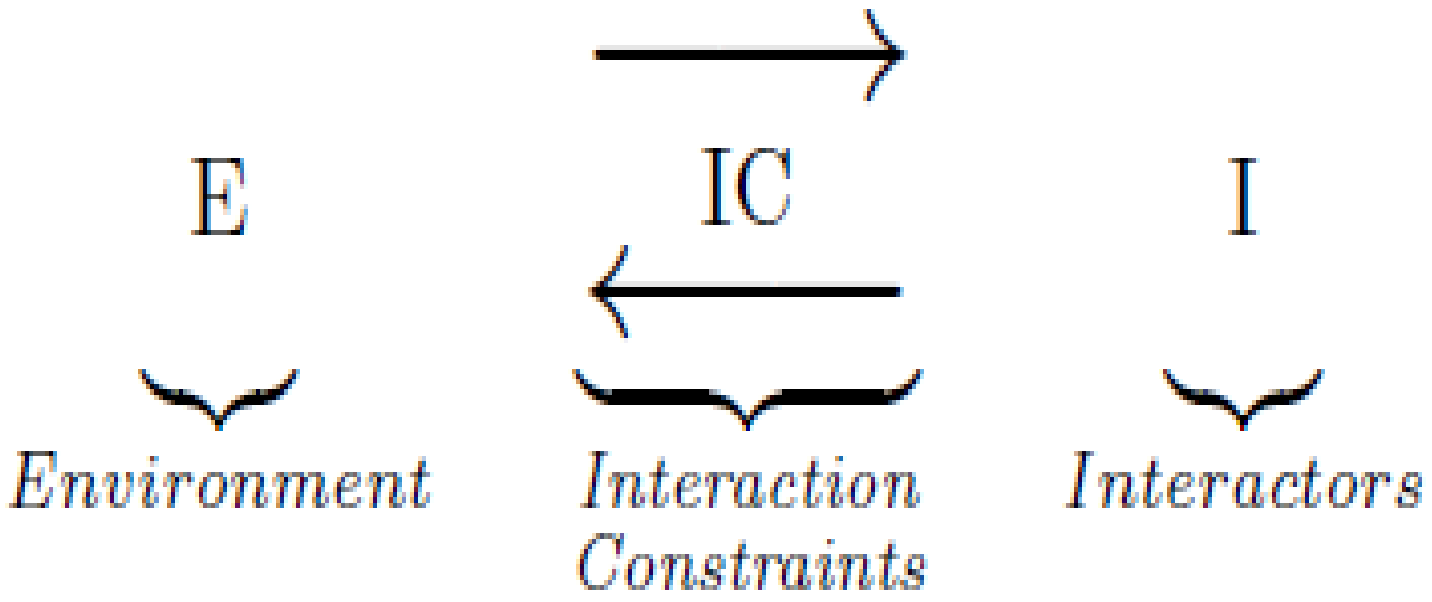
Achievement…..

Ambient Intelligence aims to enhance the way people interact with their environment to promote safety and to enrich their lives.

The achievement of Ambient Intelligence largely depends on the technology deployed (sensors and devices interconnected through networks) as well as on the intelligence of the software used for decision-making.

AmI System

$$AmI\,System = \langle E, IC, I \rangle$$



Sensing **Reasoning** **Acting**

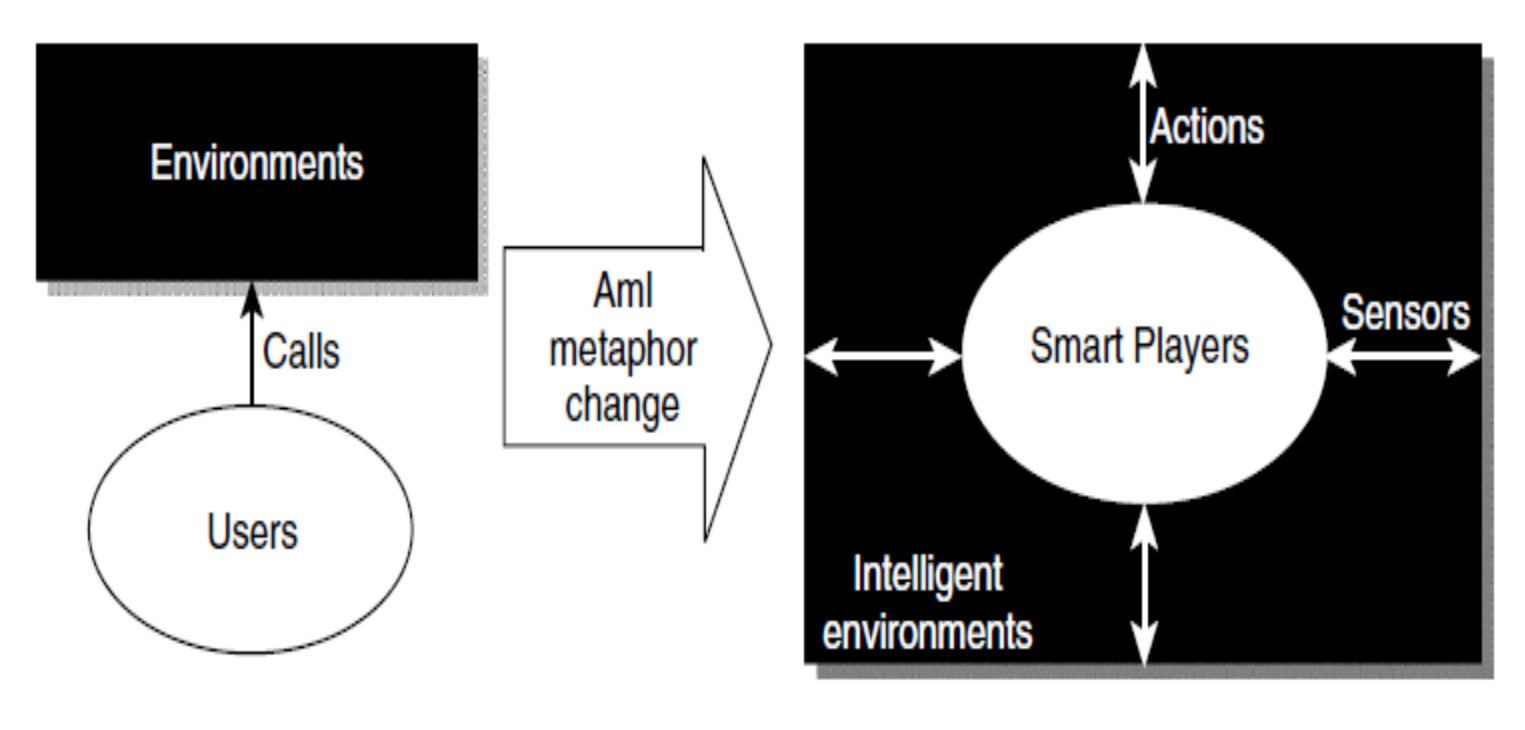
Sensing

- **Perception** is accomplished using variety of sensors.
- **Software algorithm** perceives the environment and specifies the action that can be taken to change the state of the environment.
- Sensors have been designed for **position measurement, for detection of chemicals and humidity sensing, and to determine readings for light, radiation, temperature, sound, strain, pressure, position, velocity, and direction, and physiological sensing to support health monitoring.**

Human Computer Interaction

- Explicit input must now be replaced with more human-life communication capabilities and with implicit actions.
- Technologies : motion tracking, gesture recognition, facial expression recognition and emotion recognition, speech processing, and even whistle processing facilitate natural interactions with intelligent environments.

The metaphor change of Ambient Intelligence from function orientation to goal orientation effects that a user becomes a smart player within a proactive intelligent environment



Technologies Used in AmI

- Radio Frequency Identification
- Microchip implant (human)
- Sensor
- Affective Computing
- Nanotechnology
- Biometrics

Radio Frequency Identification



- Radio-frequency identification (RFID) is the use of a wireless non-contact system that uses radio-frequency electronic fields to transfer data from a tag attached to an object.

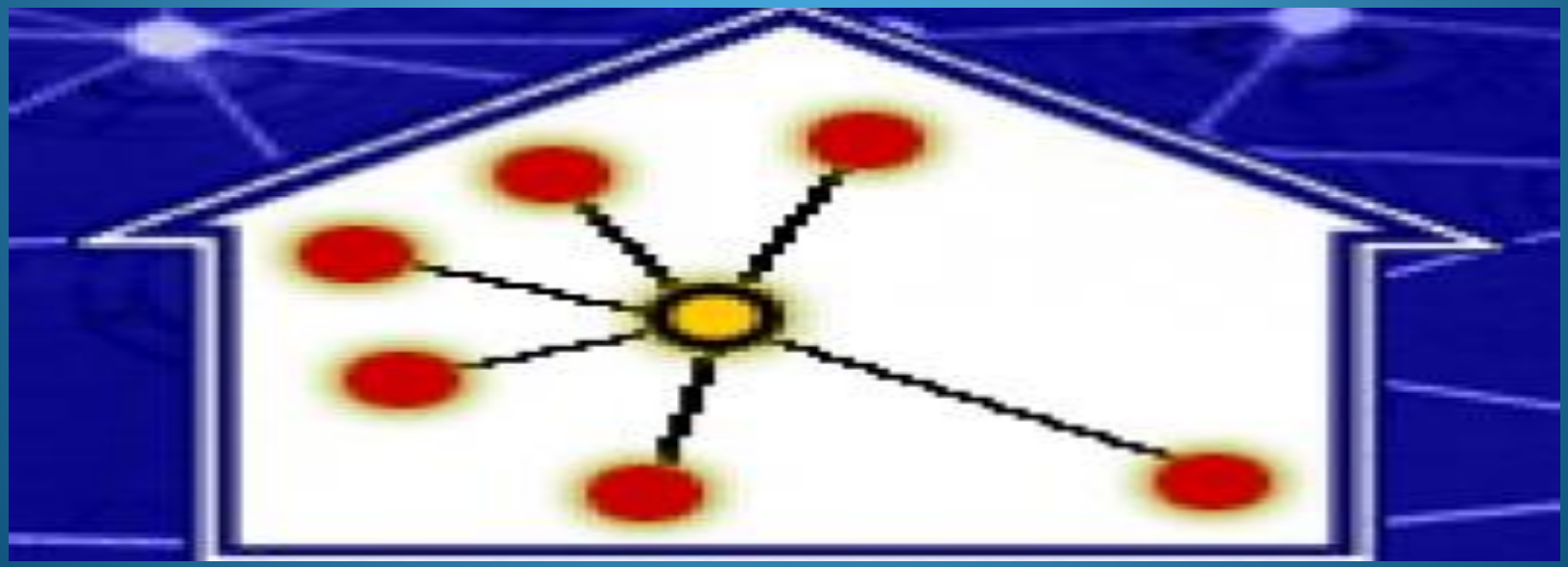
Microchip implant (human)

A human microchip implant is an integrated circuit device or RFID transponder encased in silicate glass and implanted in the body of a human being. It typically contains a unique ID number that can be linked.



Sensor

- A **sensor** is also called **detector** which is a convector that measures a physical quantity and converts it into a signal which can be read by an observer .



Affective Computing

- Affective computing is the study and development of systems and devices that can recognize human affects.
- In Affecting computing ,the machine should interpret the emotional state of humans and adapt its behavior to them, giving an appropriate response for those emotions.

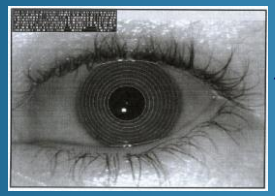
Nanotechnology- able to create many new materials and devices with a vast range of medicine , electronics.



Biometrics

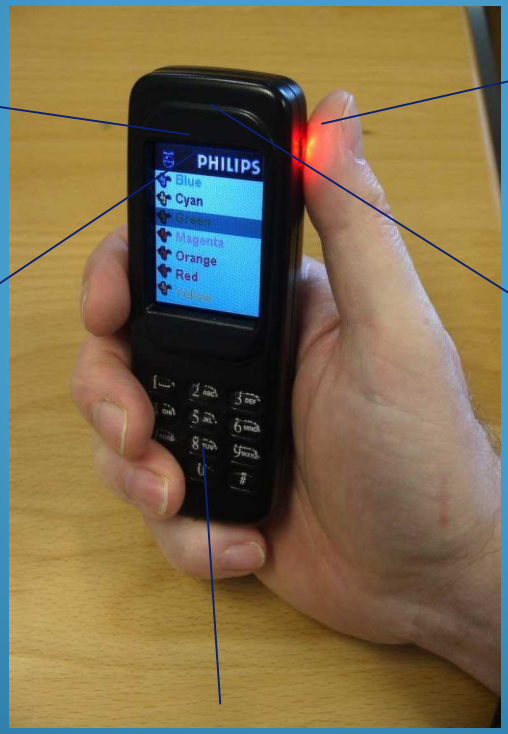
Biometrics refers to the identification of humans by their characteristics.

Iris



Robust fingerprint sensing





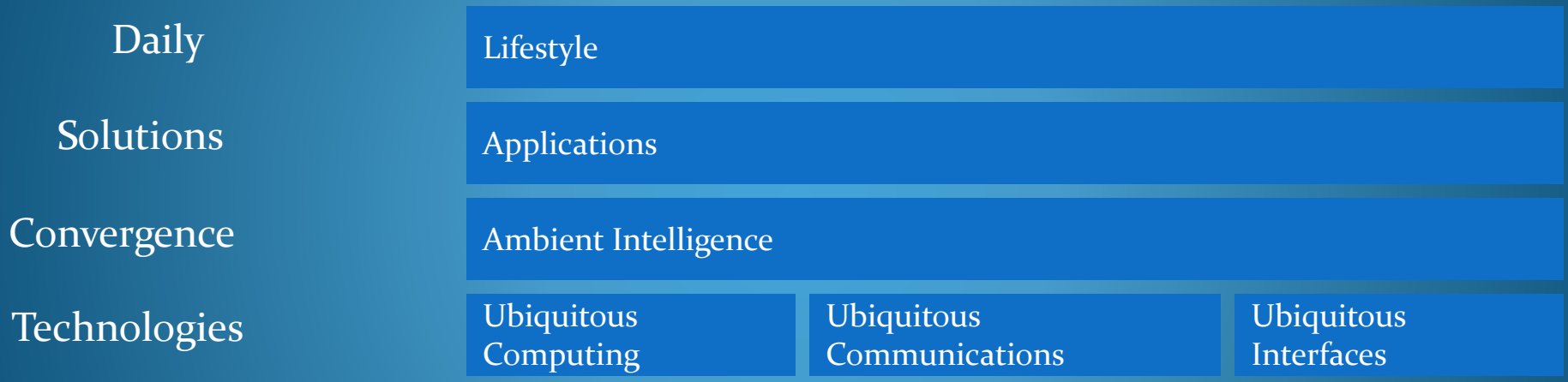


Real time Face Recognition



Ear recognition

Speaker Recognition


Daily
Solutions
Convergence
Technologies
Lifestyle
Applications
Ambient Intelligence
Ubiquitous Computing
Ubiquitous Communications
Ubiquitous Interfaces

Drawbacks

- Need high finance to install and to maintain the environment.
- Technology is to be incorporated into the daily routines of different class of people.
- Makes people lazy.

Security Risks

- AmI technologies can raise other security issues. At the sensor level, sensor reliability, handling errors, and installation errors can create security risks.

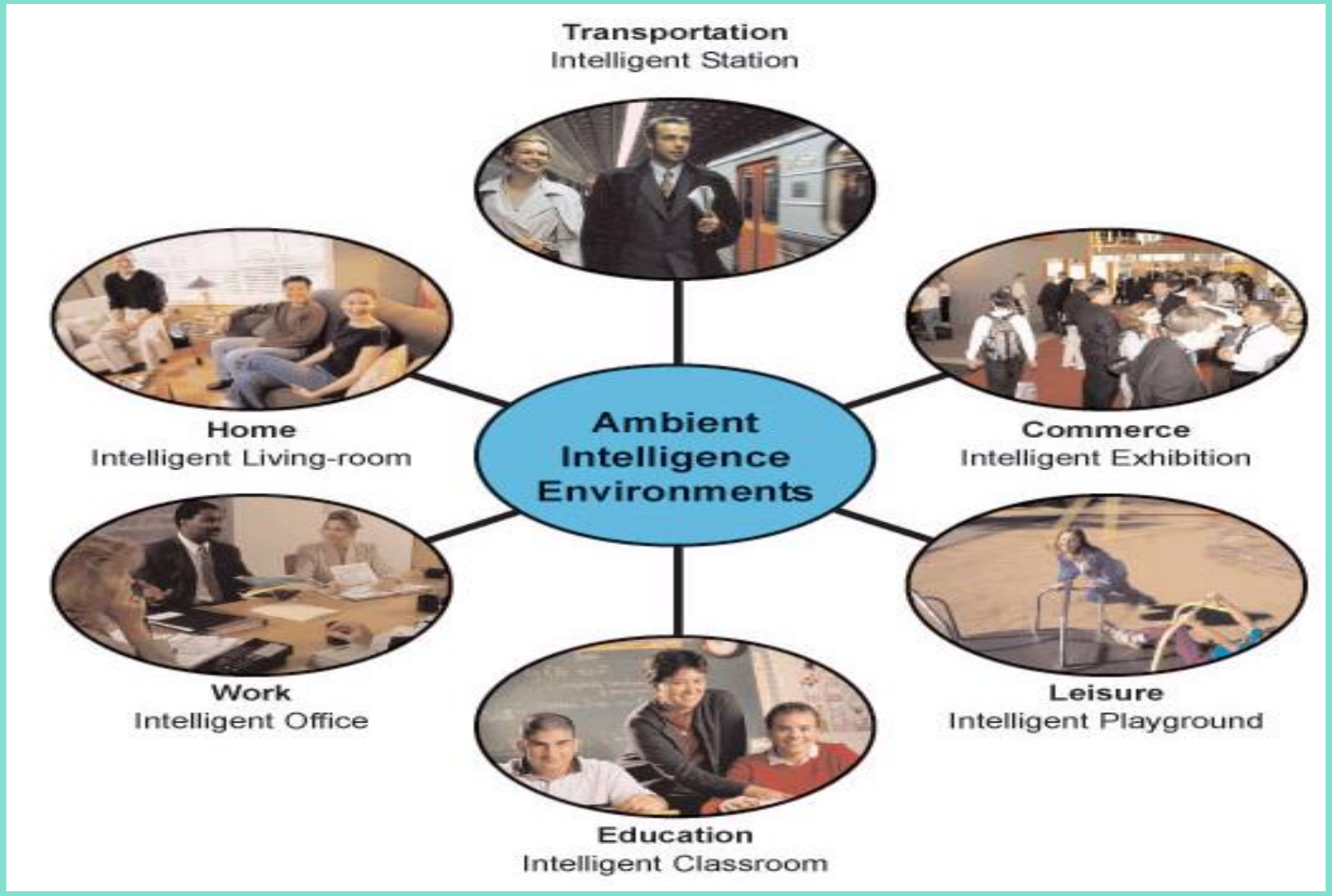
Absolutely critical for market acceptance.

Applications

- **Health-related applications**
- Public transportation sector
- Education services
- Emergency services
- Production-oriented services

Biggest adopters of AmI:

- Healthcare
- Home automation


Transportation
Intelligent Station
Home
Intelligent Living-room
Ambient Intelligence Environments
Commerce
Intelligent Exhibition
Work
Intelligent Office
Leisure
Intelligent Playground
Education
Intelligent Classroom

AmI APPLICATIONS

- Health-related applications: Hospitals can increase the efficiency of their services by monitoring patients' health and progress by performing automatic analysis of activities in their rooms. (Sensing and monitoring)

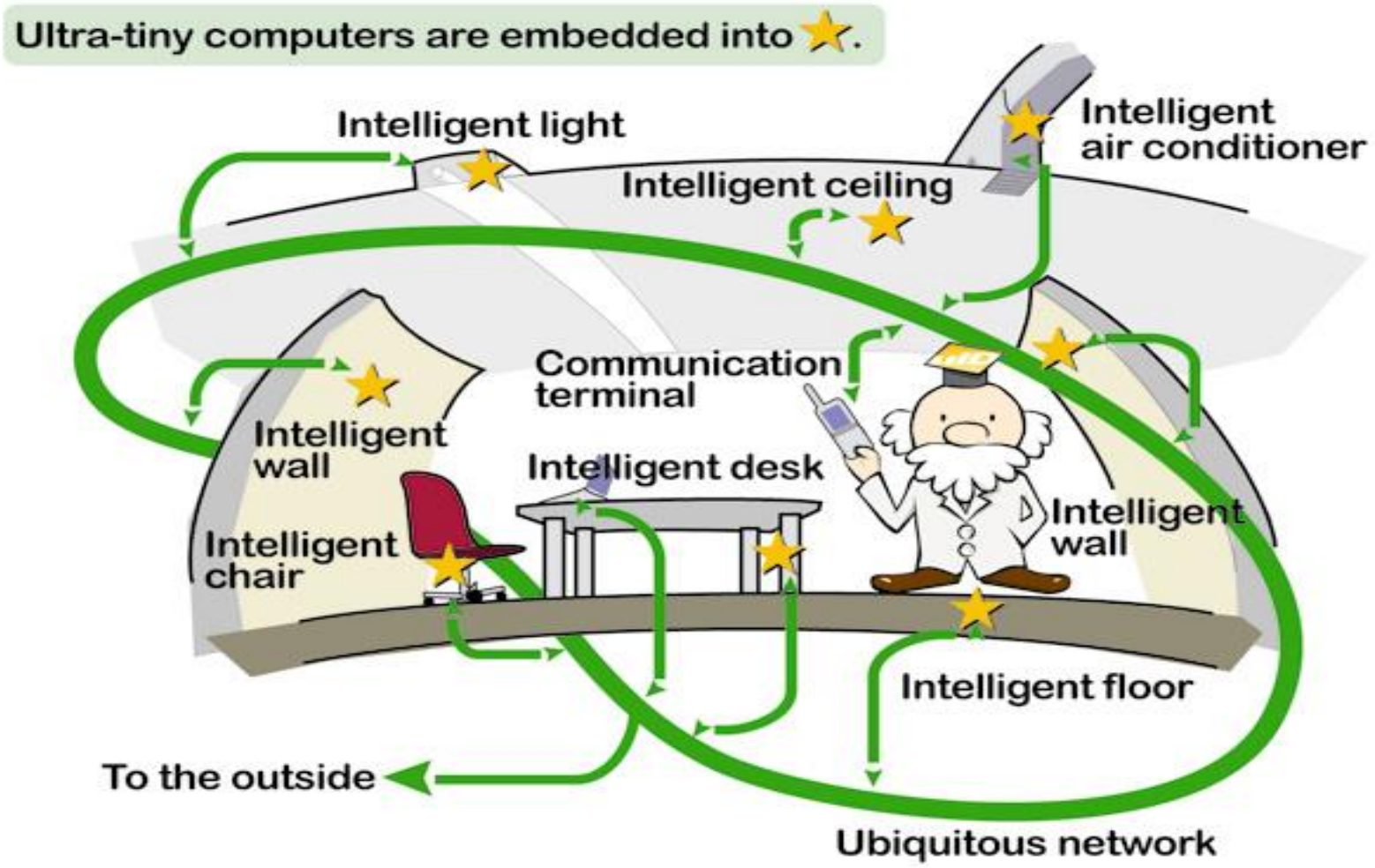
AmI APPLICATIONS

- Emergency services: Safety-related services like fire brigades can improve the reaction to a hazard by locating the place more efficiently and also by preparing the way to reach the place in connection with street services.

AmI APPLICATIONS

- Production-oriented places: Companies can use RFID sensors to tag different products and track them along the production and commercialization processes.
- It helps improving the process by providing valuable information for the company on how to react to favourable demand and unusual events like products that become unsuitable for sale.

Imagine in the future…. Intelligence Are Embedded




Based on your viewing history, you won't like the show!
The Best New Show
"Ambient intelligence apps:
Apps that tell *you* what to do"
vision mobile

AmI in Your Smart Phone…….




Thank You